Airfoil Selection for Two RC Aircraft: Payload (A) vs Maneuverability (B)

Technical report. The aerodynamic values in Section 3 are XFOIL results at Re = 200,000 and Ncrit = 9; camber and thickness for S1223, E423, FX 63-137 and Clark Y are computed from the Airfoil Tools coordinate files (the NACA 00xx values follow from the designation).

1. Mission Definitions and Assumptions

Item	Aircraft A (Payload)	Aircraft B (Maneuverability)
Span / chord	2.0 m / 0.30 m	1.2 m / 0.25 m
Mass	5.5 kg (2.5 empty + 3.0 payload)	1.5 kg
Design speed	12 m/s	15 m/s

Assumptions:

- Sizes, masses, speeds and power levels are my own design values; the brief gives none.
- Sea-level air: $\rho = 1.225$ kg/m^3, $\nu = 1.46\times10^{-5}$ m^2/s.
- The real 3D wing reaches about 0.8 × the section CL,max (rule of thumb).
- Polars are at Re = 200,000, the nearest standard value on Airfoil Tools [6], slightly below the operating Re.

Quantity	Formula	Aircraft A	Aircraft B
Wing area S	span × chord	2.0 × 0.30 = 0.60 m^2	1.2 × 0.25 = 0.30 m^2
Wing loading W/S	m·g / S	5.5 × 9.81 / 0.60 ≈ 90 N/m^2	1.5 × 9.81 / 0.30 ≈ 49 N/m^2
Aspect ratio	span^2 / S	2.0^2 / 0.60 ≈ 6.7	1.2^2 / 0.30 = 4.8
Reynolds number	V·c / ν	12 × 0.30 / 1.46×10^{-5} ≈ 247,000	15 × 0.25 / 1.46×10^{-5} ≈ 257,000
3D CL,max	0.8 × section CL,max	0.8 × 2.29 ≈ 1.83 (S1223)	0.8 × 1.11 ≈ 0.89 (NACA 0012)
Stall speed [5]	√(2·(W/S) / (ρ·CL,max))	√(2×90 / (1.225×1.83)) ≈ 9.0 m/s	√(2×49 / (1.225×0.89)) ≈ 9.5 m/s

For comparison, the Clark Y on A gives 0.8 × 1.40 ≈ 1.12 and a stall speed of about 11.5 m/s, too close to the 12 m/s design speed.

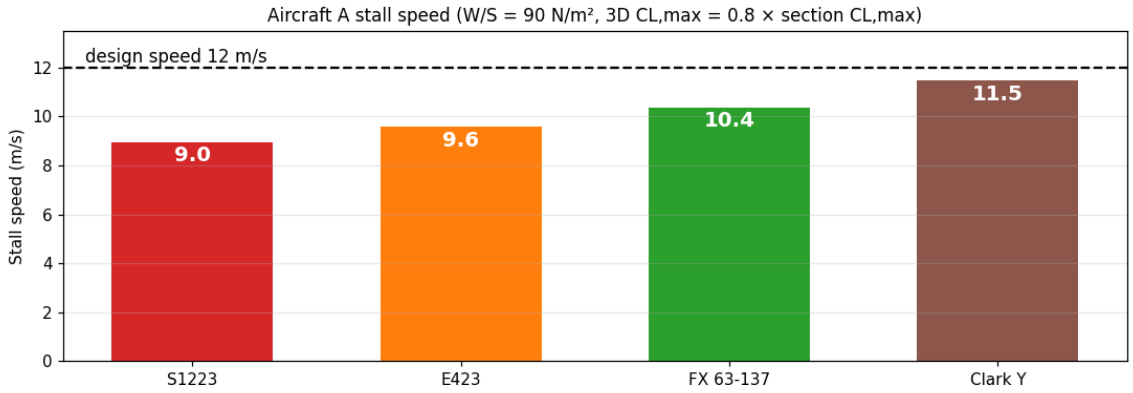


Figure 1: Stall speed of Aircraft A with each shortlisted wing section, from the CL,max in the XFOIL polars [6] and the stall-speed formula above [5].

2. Shortlisted Airfoils

Aircraft A (high lift, deep camber): Selig S1223, Eppler E423, Wortmann FX 63-137 and Clark Y (baseline).

Aircraft B (symmetric, aerobatic): NACA 0009, NACA 0012 and NACA 0015.

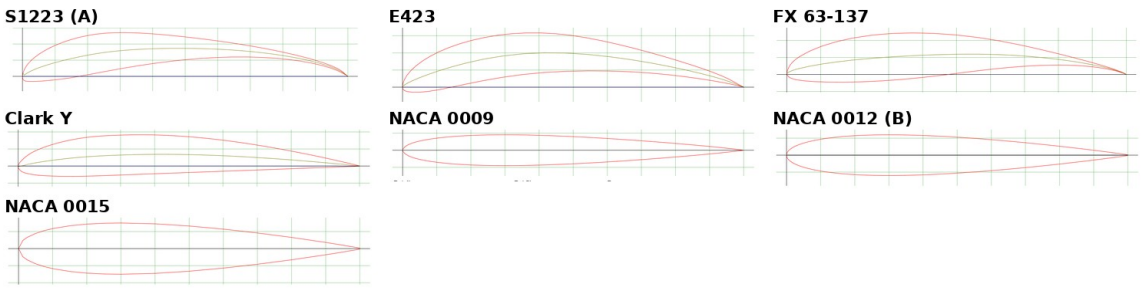


Figure 2: Candidate profiles at equal chord. Aircraft A: S1223, E423, FX 63-137, Clark Y. Aircraft B: NACA 0009, 0012, 0015. Coordinates from Airfoil Tools [6].

3. Comparison Table (Re = 200,000; XFOIL polars, Ncrit = 9)

Airfoil	Camber / thickness	CL,max	Cd min	(L/D)max	Cm	Stall
S1223	8.7% / 12.1%	2.29	0.0176	73.6	−0.27	Gradual
E423	10.0% / 12.5%	2.00	0.0192	73.7	−0.24	Very gradual

Airfoil	Camber / thickness	CL,max	Cd min	(L/D)max	Cm	Stall
FX 63-137	6.0% / 13.7%	1.71	0.0130	90.6	−0.20	Very gradual
Clark Y	3.4% / 11.7%	1.40	0.0101	73.2	−0.095	Very gentle
NACA 0009	0% / 9%	0.91	0.0089	45.6	0	Abrupt
NACA 0012	0% / 12%	1.11	0.0102	47.4	0	Moderate
NACA 0015	0% / 15%	1.21	0.0107	49.6	0	Gradual

All aerodynamic values in the table are XFOIL results from Airfoil Tools [6, 7]; camber and thickness are computed from the coordinate files [2, 6] (profiles in Figure 2). FX 63-137 shows no clear CL peak up to 17.5°. Figure 3 overlays the curves.

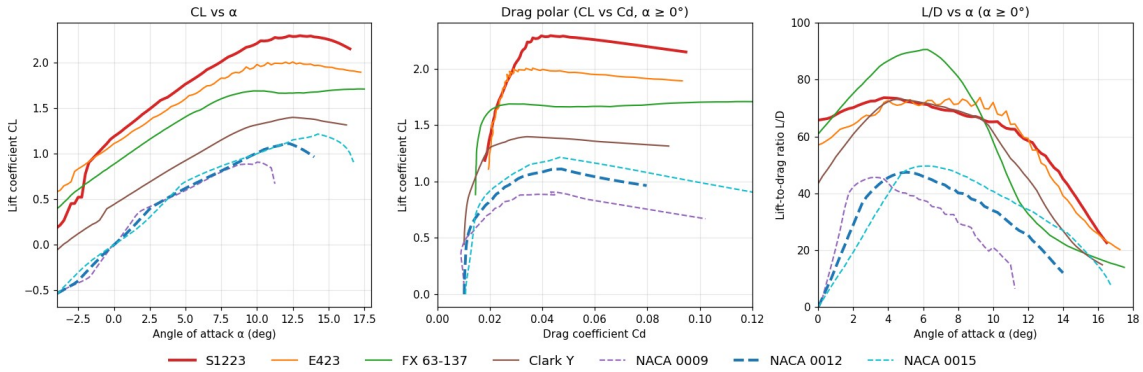


Figure 3: All seven sections at Re = 200,000, Ncrit = 9 (XFOIL via Airfoil Tools [6]). Thick lines are the selected sections (S1223 red, NACA 0012 blue); solid lines are Aircraft A candidates, dashed lines are Aircraft B candidates.

4. Selection and Justification

Aircraft A: Selig S1223

Payload capacity is set by how much lift the wing can make at an acceptable stall speed, since $W = \frac{1}{2}\rho V^2 \cdot S \cdot CL$. S1223 has the highest CL,max on the shortlist (2.29 in the XFOIL polar at Re = 200,000 [6]), which for the same wing area carries more mass or allows lower takeoff and landing speed. Its large camber gives high CL at zero angle of attack, so the fuselage can sit near level while the wing makes significant lift.

Stability trade-offs are accepted knowingly. The S1223's pitching moment is strongly nose-down (Cm ≈ −0.27), so the tail must push down to balance it, which means a larger tail and

a balance point (center of gravity) toward the front of the wing. Its very thin trailing edge is fragile, so the rear of the wing needs extra stiffening.

E423 is the closest alternative: having lower CL,max (2.00 vs 2.29) with the same peak L/D and a slightly smaller nose-down moment (−0.24 vs −0.27). Clark Y is rejected because its CL,max of 1.40 gives a stall speed of about 11.5 m/s, too close to the 12 m/s design speed.

Aircraft B: NACA 0012

Maneuverability needs roll authority, low drag so energy is retained through loops, and identical behavior upright and inverted. A symmetric section gives zero lift and zero pitching moment at $\alpha = 0$, so trim does not change between upright and inverted flight, with the same stall behavior and loading in positive and negative g.

The XFOIL polars at Re = 200,000 [6] show the trade-off. NACA 0015 has the highest CL,max (1.21) and the latest stall (14.25°), but at low CL (≈ 0.45) its drag is about 20% higher than the 0012 (0.0132 vs 0.0110), and low-CL, high-speed flight dominates loops, rolls and knife-edge passes. NACA 0012 sits in the middle of the shortlist (minimum drag 0.0102, CL,max 1.11, moderate stall) and is the standard aerobatic choice.

5. How Airfoil Geometry Affects Performance

- **Camber** raises CL at every angle of attack and lifts CL,max, but adds nose-down pitching moment and drag at low CL [4, 5].
- **Thickness** gives spar depth and a gentler stall up to about 12–15%; beyond that it adds drag.
- **Reynolds number:** below about 3×10^5 laminar separation bubbles raise drag, so low-Re sections (Selig, Eppler) outperform thin classic profiles [1].

6. Can One Airfoil Satisfy Both Missions?

Not without a serious compromise. Aircraft A needs high camber for lift and can tolerate a strong nose-down pitching moment, while Aircraft B needs zero camber, a pitching moment near zero, low drag at low CL, and the same behavior when inverted.

7. Airfoil vs. Whole-Aircraft Factors

The airfoil sets the section behavior (CL, Cd, Cm, stall type). The aircraft determines how that translates to flight:

- **Wing loading** (W/S) sets stall speed and turn radius. A's 90 N/m² versus B's 49 N/m² matters more to maneuverability than any airfoil change between 0009 and 0015.
- **Control surfaces:** aileron, elevator and rudder size and throw set the achievable rotation rates, regardless of airfoil.
- **Stability:** the tail size and the balance point (CG) decide whether A is stable; the airfoil influences this only through Cm.
- **Structure:** spar design and joints determine safe load factor. Thickness helps, but construction matters more.
- **Thrust:** B's T/W above 1.2 permits vertical climbs, which an airfoil cannot provide. A's lower T/W is acceptable for its payload mass.

8. Bonus: XFOIL Comparison (Assumptions and Method)

All values come from XFOIL 6.96 polars [3, 7] at Re = 200,000, Ncrit = 9 (a standard air-turbulence setting) and Mach = 0, taken from Airfoil Tools data files [6]. Airfoil coordinates are from the UIUC database [2]. XFOIL's CL,max may differ from wind-tunnel measurements, so the CL,max values are estimates. Results: S1223 has the highest CL,max (2.29) and the most negative Cm (−0.27); FX 63-137 has the best peak L/D (90.6); the NACA sections are symmetric about the origin. Figure 3 overlays all seven sections and highlights the two selected ones.

References

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2. UIUC Airfoil Coordinates Database, University of Illinois Urbana-Champaign (m-selig.ae.illinois.edu/ads.html).
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4. Abbott, I. H., von Doenhoff, A. E. (1959). *Theory of Wing Sections.* Dover.
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AI use: AI assistance (Claude) was used for research support, calculations, figure preparation and drafting. The data come from the cited XFOIL polars, and I am responsible for the content.